LIE ALGEBRA COHOMOLOGY

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Abstract. Aim of this manuscript is to introduce the (co)homology theory associated with a Lie algebra and to give an application in Hamiltonian dynamics. We introduce this classical theory following [Wei94, Chapter 7] via two approaches: the abstract categorical one via the notion of derived functors and the more explicit one via constructing a chain complex. We then present two fundamental results concerning semisimple Lie algebras, the so-called Whitehead Lemmas. These have direct applications in the theory of momentum maps, a central topic in Hamiltonian dynamics, providing existence and uniqueness statements. We follow [MS17, Section 5.2].

Contents

1 Lie Algebra (Co)homology **2**
2 The Chevalley–Eilenberg Complex **3**
3 The Whitehead Lemmas **6**
4 Momentum Maps **9**
References **13**

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1. Lie Algebra (Co)homology

We denote the category of all commutative rings with identity by CRing and the category of all unital left R-modules by ${}_R\mathsf{Mod}$ for some $R \in \mathsf{CRing}$.

Definition 1.1 (Lie Algebra). *Let $R \in \mathsf{CRing}$. A **Lie algebra over R** is an R-module $\mathfrak{g} \in {}_R\mathsf{Mod}$ with a bilinear product*

$$[\cdot,\cdot]\colon \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g},$$

*called the **Lie bracket**, satisfying for all $x, y, z \in \mathfrak{g}$*

(i) *(Skew-Symmetry) $[x,x] = 0$ and thus $[y,x] = -[x,y]$.*
(ii) *(Jacobi Identity) $[x,[y,z]] + [y,[z,x]] + [z,[x,y]] = 0$.*

Example 1.1 (Abelian Lie Algebra). For $R \in \mathsf{CRing}$, every $\mathfrak{g} \in {}_R\mathsf{Mod}$ is an ***abelian Lie algebra*** with $[x,y] := 0$ for all $x, y \in \mathfrak{g}$.

Example 1.2 (Lie Algebra of a Lie Group). Let M be a C^∞ finite-dimensional real manifold. Then the vector space $\mathfrak{X}(M)$ of all smooth vector fields on M is a Lie algebra with Lie bracket

$$[X,Y] := XY - YX \in \mathfrak{X}(M)$$

for all $X, Y \in \mathfrak{X}(M)$. If G is a Lie group, that is, a smooth manifold admitting a smooth group structure, then the tangent space T_eG at the identity element $e \in G$ has the canonical structure of a Lie algebra by identifying T_eG with the Lie algebra of left or right invariant vector fields on G. In fact, this construction establishes a functor from the category of smooth Lie groups to the category of Lie algebras.

Example 1.3 (Poisson Bracket). Let (M,ω) be a symplectic manifold, that is, $\omega \in \Omega^2(M)$ is closed and nondegenerate. Then $(C^\infty(M), \{\cdot,\cdot\})$ is a Lie algebra with the ***Poisson bracket***

$$\{f,g\} := \omega(X_f, X_g) \qquad \forall f,g \in C^\infty(M).$$

Moreover,

$$(C^\infty(M), \{\cdot,\cdot\}) \to (\mathfrak{X}(M), [\cdot,\cdot]) \qquad f \mapsto X_f$$

is a Lie algebra homomorphism.

Definition 1.2 (Category of Left $\mathfrak{g}$-Modules). *Let $R \in \mathsf{CRing}$ and $\mathfrak{g} \in {}_R\mathsf{LieAlg}$. The **category of left $\mathfrak{g}$-modules**, written ${}_\mathfrak{g}\mathsf{Mod}$, is defined to be the category with objects **left $\mathfrak{g}$-modules**, i.e. modules $M \in {}_R\mathsf{Mod}$ equipped with an R-bilinear product $\mathfrak{g} \times M \to M$, $(x,m) \mapsto xm$, such that*

$$[x,y]m = x(ym) - y(xm)$$

*holds for all $x, y \in \mathfrak{g}$ and $m \in M$, and **left $\mathfrak{g}$-module homomorphisms** as morphisms, i.e. morphisms $f \in \mathrm{Hom}_R(M,N)$ such that*

$$f(xm) = xf(m) \qquad \forall x \in \mathfrak{g}, m \in M.$$

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Example 1.4 (Trivial $\mathfrak{g}$-Module). A ***trivial $\mathfrak{g}$-module*** is an R-module M where $xm := 0$ for all $x \in \mathfrak{g}$ and $m \in M$.

Example 1.5 (Lie Algebra). The Lie bracket makes $\mathfrak{g}$ itself into a left $\mathfrak{g}$-module by the Jacobi identity. This module is usually called the ***adjoint representation of $\mathfrak{g}$***.

Proposition 1.1. *Let $R \in \mathsf{CRing}$ and $\mathfrak{g} \in {}_R\mathsf{LieAlg}$. Then ${}_\mathfrak{g}\mathsf{Mod}$ is an abelian category.*

Definition 1.3. *For $M \in {}_\mathfrak{g}\mathsf{Mod}$ define the **invariant submodule** by*

$$M^\mathfrak{g} := \{m \in M : xm = 0 \ \forall x \in \mathfrak{g}\} \in {}_R\mathsf{Mod}.$$

*Explicitly, we have that $M^\mathfrak{g} \cong \mathrm{Hom}_\mathfrak{g}(R, M)$, where R is viewed as a trivial $\mathfrak{g}$-module. Moreover, we define the **coinvariant** by*

$$M_\mathfrak{g} := M/\mathfrak{g}M \in {}_R\mathsf{Mod}.$$

Proposition 1.2. *Let $M \in {}_\mathfrak{g}\mathsf{Mod}$. The functor $-^\mathfrak{g}$ is right adjoint to the trivial $\mathfrak{g}$-module functor ${}_R\mathsf{Mod} \to {}_\mathfrak{g}\mathsf{Mod}$ and is therefore a left exact functor. The functor $-_\mathfrak{g}$ is left adjoint to the trivial $\mathfrak{g}$-module functor and therefore a right exact functor.*

By the preceeding Proposition, if the category ${}_\mathfrak{g}\mathsf{Mod}$ has enough projectives and injectives, we can use the concept of derived functors.

Definition 1.4 (Lie Algebra (Co)homology). *Let $M \in {}_\mathfrak{g}\mathsf{Mod}$. We write $H_*(\mathfrak{g}, M)$ for the left derived functors $L_*(-_\mathfrak{g})(M)$ of $-_\mathfrak{g}$ and call them the **homology groups of $\mathfrak{g}$ with coefficients in M**. Likewise, we write $H^*(\mathfrak{g}, M)$ for the right derived functors $R^*(-^\mathfrak{g})(M)$ of $-^\mathfrak{g}$ and call them the **cohomology groups of $\mathfrak{g}$ with coefficients in M**.*

2. The Chevalley–Eilenberg Complex

The universal envelopping algebra of a Lie algebra plays the same formal role as the group ring of a group.

Definition 2.1 (Universal Enveloping Algebra). *For $\mathfrak{g} \in {}_R\mathsf{LieAlg}$, the **universal envelopping algebra $U\mathfrak{g}$ of $\mathfrak{g}$** is defined to be the quotient of the **tensor algebra $T\mathfrak{g}$***

$$T\mathfrak{g} := \bigoplus_{n \geq 0} \mathfrak{g}^{\otimes n}$$

by the 2-sided ideal generated by the relations

$$\iota[x, y] = \iota(x)\iota(y) - \iota(y)\iota(x),$$

for all $x, y \in \mathfrak{g}$, where $\iota : \mathfrak{g} \hookrightarrow T\mathfrak{g}$ denotes inclusion.

Remark 2.1. Note that $U\mathfrak{g} \in {}_\mathfrak{g}\mathsf{Mod}$ and that $U\mathfrak{g}$ is an associative R-algebra.

Theorem 2.1 (Poincaré–Birkhoff–Witt). *If $\mathfrak{g}$ is a free R-module, then $U\mathfrak{g}$ is also a free R-module. More precisely, if (e_i) is an ordered basis of $\mathfrak{g}$, then the elements e_I with I an increasing sequence form a basis of $U\mathfrak{g}$.*

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Proposition 2.1. *The category* ${}_{\mathfrak{g}}\mathsf{Mod}$ *is naturally isomorphic to* ${}_{U\mathfrak{g}}\mathsf{Mod}$.

Proof. Let $M \in {}_R\mathsf{Mod}$ and write $E := \mathrm{End}_R(M)$ for the R-algebra of all R-module endomorphisms of M. Then there is a natural isomorphism

$$\mathrm{Hom}_{\mathsf{Lie}}(\mathfrak{g}, \mathrm{Lie}(E)) \cong \mathrm{Hom}_{R\mathsf{Alg}}(U\mathfrak{g}, E).$$

In particular, the functor is the left adjoint of the Lie algebra functor. A $\mathfrak{g}$-module is an R-module M together with a Lie algebra map $\mathfrak{g} \to \mathrm{Lie}(E)$. But a $U\mathfrak{g}$-module is an R-module M together with an associative algebra map $U\mathfrak{g} \to E$. Thus the statement follows. □

Remark 2.2. Explicitly, given $M \in {}_{\mathfrak{g}}\mathsf{Mod}$ and a monomial $x_1 \cdots x_n$ in $U\mathfrak{g}$, then the formula

$$(x_1 \cdots x_n)m := x_1(\cdots(x_n m)) \qquad \forall m \in M$$

makes M into a $U\mathfrak{g}$-module. Conversely, if M is a $U\mathfrak{g}$-module and $x \in \mathfrak{g}$, the formula $xm := \iota(x)m$ makes M into a $\mathfrak{g}$-module.

Corollary 2.1. *The category* ${}_{\mathfrak{g}}\mathsf{Mod}$ *has enough projectives and injectives.*

Definition 2.2 (Augmentation Ideal). *The unique R-algebra homomorphism $\varepsilon\colon U\mathfrak{g} \to R$ sending $\iota(\mathfrak{g})$ to zero, called the* ***augmentation****. Moreover, define the* ***augmentation ideal*** *by the formula $\mathfrak{I} := \ker \varepsilon$.*

Proposition 2.2. *Let $M \in {}_{\mathfrak{g}}\mathsf{Mod}$. Then*

$$H_*(\mathfrak{g}, M) \cong \mathrm{Tor}_*^{U\mathfrak{g}}(R, M) \qquad \textit{and} \qquad H^*(\mathfrak{g}, M) \cong \mathrm{Ext}^*_{U\mathfrak{g}}(R, M).$$

Proof. We observe that

$$R \otimes_{U\mathfrak{g}} M = (U\mathfrak{g}/\mathfrak{I}) \otimes_{U\mathfrak{g}} M \cong M/\mathfrak{I}M = M/\mathfrak{g}M = M_{\mathfrak{g}}$$

and

$$\mathrm{Hom}_{U\mathfrak{g}}(R, M) = \mathrm{Hom}_{\mathfrak{g}}(R, M) = M^{\mathfrak{g}}.$$

Thus the statement follows by observing that in order to show that two derived functors are isomorphic it is enough to show that the underlying functors are isomorphic. □

Definition 2.3 (Chevalley–Eilenberg Complex). *Let $R \in \mathsf{CRing}$ and $\mathfrak{g} \in {}_R\mathsf{LieAlg}$ which is free as an R-module. Denote by $U\mathfrak{g}$ the universal envelopping algebra of $\mathfrak{g}$. Define a non-negative chain complex $(C_*(\mathfrak{g}), \partial_*) \in \mathrm{Ch}_{\geq 0}({}_{U\mathfrak{g}}\mathsf{Mod})$ by*

$$C_n(\mathfrak{g}) := U\mathfrak{g} \otimes_R \Lambda^n \mathfrak{g}$$

for all $n \geq 0$ and $\partial_n\colon C_n(\mathfrak{g}) \to C_{n-1}(\mathfrak{g})$ is given by

$$\partial_n(u \otimes x_1 \wedge \cdots \wedge x_n) := \begin{cases} ux_1 & n = 1, \\ \theta_1 + \theta_2 & n > 1, \end{cases}$$

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where

$$\theta_1 := \sum_{i=1}^{n} (-1)^{i+1} u x_i \otimes x_1 \wedge \cdots \wedge \hat{x}_i \wedge \cdots \wedge x_n,$$

and

$$\theta_2 := \sum_{1 \le i < j \le n} (-1)^{i+j} u \otimes [x_i, x_j] \wedge x_1 \wedge \cdots \wedge \hat{x}_i \wedge \cdots \wedge \hat{x}_j \wedge \cdots \wedge x_n.$$

Remark 2.3. It is by no means obvious, that $\partial_n \circ \partial_{n+1} = 0$ holds for the Chevalley-Eilenberg complex 2.3. However, it is a tedious computation, and we will only demonstrate the case $n = 1$. In this case

$$\begin{aligned}(\partial_1 \circ \partial_2)(u \otimes x \wedge y) &= \partial_1(ux \otimes y - uy \otimes x - u \otimes [x, y]) \\ &= u(xy - yx) - u[x, y] \\ &= 0,\end{aligned}$$

for all $u \in U\mathfrak{g}$ and $x, y \in \mathfrak{g}$. The definition of the boundary operator has great similarities with the invariant formula for the exterior derivative. Indeed, let M be a smooth manifold and $\omega \in \Omega^k(M)$. Then for any $X_1, \ldots, X_{k+1} \in \mathfrak{X}(M)$ we have that

$$\begin{aligned}d\omega(X_1, \ldots, X_{k+1}) = &\sum_{1 \le i \le k+1} (-1)^{i-1} X_i \left(\omega(X_1, \ldots, \hat{X}_i, \ldots, X_{k+1})\right) \\ &+ \sum_{1 \le i < j \le k+1} (-1)^{i+j} \omega\left([X_i, X_j], X_1, \ldots, \hat{X}_i, \ldots, \hat{X}_j, \ldots, X_{k+1}\right).\end{aligned}$$

Theorem 2.2 (Chevalley–Eilenberg). *If $M \in \mathsf{Mod}_{\mathfrak{g}}$, then the homology modules $H_*(\mathfrak{g}, M)$ are the homology of the chain complex $(M \otimes_R \Lambda^* \mathfrak{g}, \partial_*)$. Likewise, if $M \in {}_{\mathfrak{g}}\mathsf{Mod}$, then the cohomology modules $H^*(\mathfrak{g}, M)$ are the cohomology modules of the cochain complex*

$$\mathrm{Hom}_{\mathfrak{g}}(C_*(\mathfrak{g}), M) \cong \mathrm{Hom}_R(\Lambda^* \mathfrak{g}, M) \cong \mathrm{Alt}^*_R(\mathfrak{g}, M)$$

with coboundary operator

$$\begin{aligned}d^n f(x_1, \ldots, x_n) = &\sum_{i=1}^{n} (-1)^{i+1} x_i f(x_1, \ldots, \hat{x}_i, \ldots, x_n) \\ &+ \sum_{1 \le i < j \le n} (-1)^{i+j} f([x_i, x_j], x_1, \ldots, \hat{x}_i, \ldots, \hat{x}_j, \ldots, x_n),\end{aligned}$$

for $n > 1$ and

$$d^1 f(x_1) = x_1 f(1).$$

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Example 2.1 ($H^3(\mathfrak{sl}_2(k), k)$). Let k be a field of characteristic not equal to two and consider the ***special linear Lie algebra over k***, that is, the algebra of all $A \in k^{2\times 2}$ such that $\operatorname{tr} A = 0$. This is the three dimensional Lie algebra $\mathfrak{sl}_2(k)$ with ordered basis

$$e_1 := \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \qquad e_2 := \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \qquad e_3 := \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Hence the crucial portion of the Eilenberg-Chevalley cochain complex is given by

$$\ldots \longrightarrow \operatorname{Hom}_k(\Lambda^2\mathfrak{sl}_2(k), k) \xrightarrow{d} \operatorname{Hom}_k(\Lambda^3\mathfrak{sl}_2(k), k) \longrightarrow 0.$$

We compute

$$\begin{aligned} df(e_1, e_2, e_3) =& e_1 f(e_2, e_3) - e_2 f(e_1, e_3) + e_3 f(e_1, e_2) \\ & - f([e_1, e_2], e_3) + f([e_1, e_3], e_2) - f([e_2, e_3], e_1) \\ =& -2f(e_2, e_3) - 2f(e_3, e_2) - f(e_1, e_1) \\ =& -2f(e_2, e_3) + 2f(e_2, e_3) \\ =& 0, \end{aligned}$$

since k is interpreted as a trivial $\mathfrak{sl}_2$-module. Hence

$$H^3(\mathfrak{sl}_2(k), k) \cong \operatorname{Hom}_R(\Lambda^3\mathfrak{sl}_2(k), k) \cong k.$$

Proof sketch of Theorem 2.2. The main part of the proof consist of showing that

$$C_*(\mathfrak{g}) \xrightarrow{\varepsilon} R$$

is a projective resolution of the $\mathfrak{g}$-module R. It is enough to show that $H_n(C_*(\mathfrak{g})) = 0$ for all $n \neq 0$. As $\mathfrak{g}$ is assumed to be free as an R-module, one can invoke the Poincaré–Birkhoff–Witt Theorem 2.1 to obtain a filtration of $C_*(\mathfrak{g})$ by R-submodules. Using this filtration, we obtain a convergent spectral sequence and the statement follows by analysing this spectral sequence. □

3. The Whitehead Lemmas

Definition 3.1 (Semisimple Lie Algebra). *A finite-dimensional Lie algebra $\mathfrak{g}$ over a field k with characteristic zero is said to be **semisimple**, if $\mathfrak{g}$ admits no nonzero abelian ideals.*

Example 3.1 ($\mathfrak{sl}_n(k)$). The special linear Lie algebra

$$\mathfrak{sl}_n(k) := \{A \in k^{n\times n} : \operatorname{tr} A = 0\}$$

for a field k of characteristic zero is a semisimple Lie algebra for $n \geq 2$.

Theorem 3.1 (Whitehead's First Lemma). *Let k be a field of characteristic zero and $\mathfrak{g} \in {}_k\mathsf{LieAlg}$ semisimple. Then for any finite-dimensional $M \in {}_{\mathfrak{g}}\mathsf{Mod}$ we have that*

$$H^1(\mathfrak{g}, M) = 0.$$

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Proof. We proceed by induction on $\dim M$. If M is simple, then either $M = k$ and $H^1(\mathfrak{g}, k) = \mathfrak{g}/[\mathfrak{g}, \mathfrak{g}] = 0$ or $M \neq k$ and $H^*(\mathfrak{g}, M) = 0$. Indeed, in the latter case, one can show that

$$H^i(\mathfrak{g}, M) = 0 \qquad \forall i$$

if $\mathfrak{g}$ is semisimple and $M \neq k$ is simple as a $\mathfrak{g}$-module. This basically follows from Proposition 2.2 together with the simplicity of M. Otherwise, M contains a proper submodule L. By induction, $H^1(\mathfrak{g}, L) = H^1(\mathfrak{g}, M/L) = 0$ and the result follows by considering the cohomology exact sequence

$$\ldots \longrightarrow H^1(\mathfrak{g}, L) \longrightarrow H^1(\mathfrak{g}, M) \longrightarrow H^1(\mathfrak{g}, M/L) \longrightarrow \ldots$$

□

Definition 3.2 (Derivation). *Let $M \in {}_{\mathfrak{g}}\mathsf{Mod}$. A **derivation from $\mathfrak{g}$ into M** is an R-linear map $D: \mathfrak{g} \to M$ satisfying the Leibniz formula*

$$D[x, y] = x(Dy) - y(Dx) \qquad \forall x, y \in \mathfrak{g}.$$

The R-submodule of $\mathrm{Hom}_R(\mathfrak{g}, M)$ of all such derivations is denoted by $\mathcal{D}(\mathfrak{g}, M)$.

Note that $\mathcal{D}(\mathfrak{g}, \mathfrak{g})$ is a derivation algebra.

Example 3.2 (Inner Derivation). If $m \in M$, define $D_m(x) := xm$ for all $x \in \mathfrak{g}$. Then $D_m \in \mathcal{D}(\mathfrak{g}, M)$ and every D_m is called an ***inner derivation***. The R-submodule of $\mathcal{D}(\mathfrak{g}, M)$ of all inner derivations is denoted by $\mathcal{D}_{\mathrm{Inn}}(\mathfrak{g}, M)$.

Proposition 3.1. *Let $M \in {}_{\mathfrak{g}}\mathsf{Mod}$. Then*

$$H^1(\mathfrak{g}, M) \cong \mathcal{D}(\mathfrak{g}, M)/\mathcal{D}_{\mathrm{Inn}}(\mathfrak{g}, M).$$

Proof. If $\varphi: \mathfrak{I} \to M$ extends to a $\mathfrak{g}$-map $U\mathfrak{g} \to M$ sending 1 to $m \in M$, then

$$D_\varphi(x) = \varphi(x \cdot 1) = xm = D_m(x) \qquad \forall x \in \mathfrak{g},$$

where $D_\varphi: \mathfrak{g} \to M$ is defined by $D_\varphi(x) := \varphi(\iota(x))$. This implies that D_φ is an inner derivation. Observe that $\varphi \mapsto D_\varphi$ establishes a natural isomorphism of R-modules

$$\mathrm{Hom}_{\mathfrak{g}}(\mathfrak{I}, M) \cong \mathcal{D}(\mathfrak{g}, M).$$

Thus the image of

$$M \to \mathcal{D}(\mathfrak{g}, M)$$

is the submodule of inner derivations. □

Corollary 3.1. *Let k be a field of characteristic zero and $\mathfrak{g} \in {}_k\mathsf{LieAlg}$ semisimple. Then for any finite-dimensional $M \in {}_{\mathfrak{g}}\mathsf{Mod}$, every derivation from $\mathfrak{g}$ into M is an inner derivation.*

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Definition 3.3 (Extension of Lie Algebras). *An **extension of Lie algebras** is a short exact sequence*

$$0 \longrightarrow M \longrightarrow \mathfrak{e} \xrightarrow{\pi} \mathfrak{g} \longrightarrow 0,$$

where M is an abelian Lie algebra. There is also a notion of equivalence of two extensions of $\mathfrak{g}$ by M. We denote the set of all such equivalence classes by $\mathcal{E}(\mathfrak{g}, M)$.

Proposition 3.2. *Let $M \in {}_{\mathfrak{g}}\mathsf{Mod}$. Then*

$$\mathcal{E}(\mathfrak{g}, M) \xleftrightarrow{1:1} H^2(\mathfrak{g}, M).$$

Proof. This follows from the so-called ***Hochschild–Serre spectral sequence***. For every ideal $\mathfrak{h}$ of a Lie algebra $\mathfrak{g}$, there are two convergent first quadrant spectral sequences

$$E^2_{pq} = H_p(\mathfrak{g}/\mathfrak{h}, H_q(\mathfrak{h}, M)) \qquad \Rightarrow \qquad H_{p+q}(\mathfrak{g}, M),$$

and

$$E_2^{pq} = H^p(\mathfrak{g}/\mathfrak{h}, H^q(\mathfrak{h}, M)) \qquad \Rightarrow \qquad H^{p+q}(\mathfrak{g}, M),$$

coming from the exact sequence

$$0 \longrightarrow \mathfrak{h} \longrightarrow \mathfrak{g} \longrightarrow \mathfrak{g}/\mathfrak{h} \longrightarrow 0.$$

□

Theorem 3.2 (Whitehead's Second Lemma). *Let k be a field of characteristic zero and $\mathfrak{g} \in {}_k\mathsf{LieAlg}$ semisimple. Then for any finite-dimensional $M \in {}_{\mathfrak{g}}\mathsf{Mod}$ we have that*

$$H^2(\mathfrak{g}, M) = 0.$$

Remark 3.1. There cannot be a third Whitehead lemma, since

$$H^3(\mathfrak{sl}_2(k), k) \cong k,$$

by Example 2.1.

Proof. As the cohomology functor H^* commutes with direct sums, and M is a direct sum of simple $\mathfrak{g}$-modules, we may assume that M is simple. Indeed, this follows from Weyl's theorem (a corollary of Whitehead's first Lemma), stating that every finite-dimensional $\mathfrak{g}$-module M is completely reducible or equivalently, can be written as a direct sum of simple $\mathfrak{g}$-modules. If $M \neq k$, we immediately obtain $H^2(\mathfrak{g}, M) = 0$. Hence it suffices to show that $H^2(\mathfrak{g}, k) = 0$, that is, every Lie algebra extension

$$0 \longrightarrow k \longrightarrow \mathfrak{e} \xrightarrow{\pi} \mathfrak{g} \longrightarrow 0$$

splits. This follows again from applying Weyl's Theorem. □

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4. Momentum Maps

Let $\psi: G \times M \to M$ be a smooth left action of a Lie group G on a smooth manifold M and denote by $\mathfrak{g} := T_eG \cong \mathfrak{X}_L(G)$ the corresponding Lie algebra. Every $\xi \in \mathfrak{g}$ determines a smooth global flow on M by $(t, x) \mapsto \psi_{\exp(-t\xi)}(x)$, where

$$\exp: \mathfrak{g} \to G, \qquad \exp(\xi) := \gamma_\xi(1)$$

denotes the exponential map and γ_ξ is the integral curve of the left-invariant vector field X_ξ starting at e with $\dot{\gamma}_\xi(0) = \xi$. Note that if there exists a bi-invariant Riemannian metric on G, then this exponential map coincides with the exponential map of the associated Levi–Civita connection at the identity. Define $\widehat{\xi} \in \mathfrak{X}(M)$ to be the infinitesimal generator of this flow, that is,

$$\widehat{\xi}_x = \frac{d}{dt}\bigg|_{t=0} \psi_{\exp(-t\xi)}(x) \qquad \forall x \in M.$$

Then

$$(\mathfrak{g}, [\cdot,\cdot]) \to (\mathfrak{X}(M), [\cdot,\cdot]), \qquad \xi \mapsto \widehat{\xi}$$

is a Lie algebra homomorphism.

Definition 4.1 (Weakly Hamiltonian Action). *Let (M, ω) be a symplectic manifold. A smooth action $\psi: G \times M \to M$ of a Lie group G is said to be **weakly Hamiltonian**, if $\psi_g^* \omega = \omega$ for all $g \in G$ and there exists a linear map*

$$\mu: \mathfrak{g} \to C^\infty(M),$$

*called a **momentum map**, such that the diagram*

$$\begin{array}{ccc} C^\infty(M) & \xrightarrow{f \mapsto X_f} & \mathfrak{X}(M, \omega) \\ & {\scriptstyle \mu} \nwarrow \quad \nearrow {\scriptstyle \xi \mapsto \widehat{\xi}} & \\ & \mathfrak{g} & \end{array}$$

commutes.

Definition 4.2. *A weakly Hamiltonian action of a Lie group G on a symplectic manifold (M, ω) is called*

- ***Hamiltonian**, if the momentum map $\mu: \mathfrak{g} \to C^\infty(M)$ is G-equivariant with respect to the adjoint action of G on its associated Lie algebra $\mathfrak{g}$ and the induced action of G on $C^\infty(M)$, that is*

  $$\mu(\mathrm{Ad}_{g^{-1}}(\xi)) = \mu(\xi) \circ \psi_g \qquad \forall g \in G, \xi \in \mathfrak{g},$$

  where

  $$\mathrm{Ad}_{g^{-1}}(\xi) := \frac{d}{dt}\bigg|_{t=0} g^{-1} \exp(t\xi) g.$$

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- ***Poisson**, if the associated momentum map*

$$\mu\colon (\mathfrak{g}, [\cdot,\cdot]) \to (C^\infty(M), \{\cdot,\cdot\})$$

is a Lie algebra homomorphism.

Remark 4.1. By [MS17, Lemma 5.2.1], every weakly Hamiltonian action by a connected Lie group is Hamiltonian if and only if it is Poisson.

Theorem 4.1 (Uniqueness of Momentum Maps for Poisson Actions). *Let μ and $\widetilde{\mu}$ be two momentum maps for a Poisson G-action on a connected symplectic manifold. If we have that $H^1(\mathfrak{g};\mathbb{R}) = 0$, then $\mu = \widetilde{\mu}$.*

Proof. By assumption there exists $\sigma \in \mathfrak{g}^\vee$ such that

$$\mu(\xi) - \widetilde{\mu}(\xi) = \sigma(\xi) \qquad \forall \xi \in \mathfrak{g}.$$

Since both μ and $\widetilde{\mu}$ are Lie algebra homomorphisms, we have that $d\sigma = 0$. Indeed, for all $\xi, \eta \in \mathfrak{g}$ we compute

$$d\sigma(\xi,\eta) = \sigma([\eta,\xi]) = \mu([\eta,\xi]) - \widetilde{\mu}([\eta,\xi]) = \{\mu(\eta),\mu(\xi)\} - \{\widetilde{\mu}(\eta),\widetilde{\mu}(\xi)\} = 0.$$

Thus $\sigma \in H^1(\mathfrak{g};\mathbb{R}) = 0$, implying $\sigma = 0$ and the statement follows. □

Theorem 4.2 (Existence of Poisson Actions). *Suppose we are given a weakly Hamiltonian G-action on a connected symplectic manifold (M,ω). If $H^2(\mathfrak{g};\mathbb{R}) = 0$, then the action is Poisson.*

Proof. For $\xi, \eta \in \mathfrak{g}$ we compute

$$X_{\mu([\xi,\eta])} = \widehat{[\xi,\eta]} = [\hat{\xi},\hat{\eta}] = [X_{\mu(\xi)}, X_{\mu(\eta)}] = X_{\{\mu(\xi),\mu(\eta)\}}.$$

Thus by connectedness of M there exists $\tau \in \Lambda^2\mathfrak{g}^\vee$ such that

$$\{\mu(\xi),\mu(\eta)\} - \mu([\xi,\eta]) = \tau(\xi,\eta) \qquad \forall \xi,\eta \in \mathfrak{g}.$$

Invoking the Jacobi identity for the Lie as well as the Poisson bracket, we conclude that $d\tau = 0$ and so $\tau \in H^2(\mathfrak{g};\mathbb{R}) = 0$. See also [MS17, Lemma 5.2.3]. Hence there exists $\sigma \in H^1(\mathfrak{g};\mathbb{R})$ such that $\tau = d\sigma$. The momentum map

$$\mathfrak{g} \to C^\infty(M), \qquad \xi \mapsto \mu(\xi) - \sigma(\xi)$$

is a Lie algebra homomorphism. □

Corollary 4.1. *Let G be a Lie group with connected semisimple Lie algebra. Then every weakly Hamiltonian G-action on a connected symplectic manifold is Hamiltonian and Poisson with a unique momentum map.*

Remark 4.2. By [Sil08, Theorem 26.1], we have that

$$H^k(\mathfrak{g};\mathbb{R}) \cong H^k_{\mathrm{dR}}(G)$$

for the Lie algebra $\mathfrak{g}$ of a compact and connected Lie group G.

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Lemma 4.1 (Momentum Lemma). *Let $\psi\colon G \times M \to M$ be a smooth Lie group action on an exact symplectic manifold (M, λ) such that $\psi_g^*\lambda = \lambda$ for all $g \in G$ holds. Then the action is Hamiltonian and Poisson with momentum map*

$$\mu(\xi) = i_{\widehat{\xi}}\lambda, \qquad \forall \xi \in \mathfrak{g}.$$

Proof. We show the result in four steps. Obviously, $\psi_g^* d\lambda = d\lambda$ for all $g \in G$.

Step 1: ψ is a weakly Hamiltonian action. Let $\xi \in \mathfrak{g}$. We compute

$$\begin{aligned} i_{\widehat{\xi}} d\lambda &= L_{\widehat{\xi}}\lambda - d i_{\widehat{\xi}}\lambda \\ &= \frac{d}{dt}\bigg|_{t=0} \psi_{\exp(-t\xi)}^*\lambda - d\mu(\xi) \\ &= \frac{d}{dt}\bigg|_{t=0} \lambda - d\mu(\xi) \\ &= -d\mu(\xi). \end{aligned}$$

Step 2: $\psi_g^*\widehat{\xi} = \widehat{\mathrm{Ad}_{g^{-1}}(\xi)}$ for all $g \in G$ and $\xi \in \mathfrak{g}$. We have a commutative diagram

$$\begin{array}{ccc} \mathfrak{g} & \xrightarrow{\mathrm{Ad}_{g^{-1}}} & \mathfrak{g} \\ {\scriptstyle \exp}\downarrow & & \downarrow{\scriptstyle \exp} \\ G & \xrightarrow[C_{g^{-1}}]{} & G, \end{array}$$

where $C_{g^{-1}}(h) = g^{-1}hg$ denotes the conjugation action on G. Let $g \in G$. Then we compute

$$\begin{aligned} \psi_g^*\widehat{\xi} &= D\psi_{g^{-1}} \circ \widehat{\xi} \circ \psi_g \\ &= D\psi_{g^{-1}} \circ \frac{d}{dt}\bigg|_{t=0} \psi_{\exp(-t\xi)} \circ \psi_g \\ &= \frac{d}{dt}\bigg|_{t=0} \psi_{g^{-1}} \circ \psi_{\exp(-t\xi)} \circ \psi_g \\ &= \frac{d}{dt}\bigg|_{t=0} \psi_{g^{-1}\exp(-t\xi)g} \\ &= \frac{d}{dt}\bigg|_{t=0} \psi_{\exp(-t\,\mathrm{Ad}_{g^{-1}}(\xi))} \\ &= \widehat{\mathrm{Ad}_{g^{-1}}(\xi)}. \end{aligned}$$

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Step 3: ψ is a Hamiltonian action. Using Step 2, we compute

$$\begin{aligned}\mu(\mathrm{Ad}_{g^{-1}}(\xi)) &= i_{\widehat{\mathrm{Ad}_{g^{-1}}(\xi)}}\lambda \\ &= i_{\psi_g^*\widehat{\xi}}\lambda \\ &= \lambda(D\psi_{g^{-1}} \circ \widehat{\xi} \circ \psi_g) \\ &= \psi_{-g}^*\lambda(\widehat{\xi} \circ \psi_g) \\ &= i_{\widehat{\xi}}\lambda \circ \psi_g \\ &= \mu(\xi) \circ \psi_g\end{aligned}$$

for all $g \in G$ and $\xi \in \mathfrak{g}$.

Step 4: ψ is a Poisson action. For $\xi, \eta \in \mathfrak{g}$ we compute

$$\begin{aligned}\mu[\xi,\eta] &= i_{\widehat{[\xi,\eta]}}\lambda \\ &= i_{[\widehat{\xi},\widehat{\eta}]}\lambda \\ &= L_{\widehat{\xi}}i_{\widehat{\eta}}\lambda - i_{\widehat{\eta}}L_{\widehat{\xi}}\lambda \\ &= L_{\widehat{\xi}}i_{\widehat{\eta}}\lambda \\ &= L_{\widehat{\xi}}\mu(\eta) \\ &= \widehat{\xi}\mu(\eta) \\ &= X_{\mu(\xi)}\mu(\eta) \\ &= \{\mu(\xi), \mu(\eta)\}.\end{aligned}$$

□

Example 4.1 (Cotangent Lift). Let M be a smooth manifold and $\varphi \in \mathrm{Diff}(M)$. Define a map $D\varphi^\dagger \colon T^\vee M \to T^\vee M$, called the ***cotangent lift of φ***, by

$$D\varphi^\dagger(q,p)(v) := p\left(D\varphi^{-1}(v)\right) \qquad \forall v \in T_{\varphi(q)}M.$$

Then one checks that $(D\varphi^\dagger)^*\lambda = \lambda$ for the canonical Liouville form λ on $T^\vee M$. Thus if we have a smooth Lie group action $\psi \colon G \times M \to M$, we get an induced action

$$\Psi \colon G \times T^\vee M \to T^\vee M, \qquad \Psi_g := D\psi_g^\dagger.$$

Then by the Momentum Lemma 4.1 we conclude that Ψ is Hamiltonian and Poisson with momentum map

$$\begin{aligned}\mu(\xi)(q,p) &= \lambda_{(q,p)}\left(\widehat{\xi}(q,p)\right) \\ &= p\left(D\pi\left(\widehat{\xi}(q,p)\right)\right) \\ &= p\left(\left.\frac{d}{dt}\right|_{t=0} \pi\left(\Psi_{\exp(-t\xi)}(q,p)\right)\right)\end{aligned}$$

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$$= p\left(\left.\frac{d}{dt}\right|_{t=0} \psi_{\exp(-t\xi)}(\pi(q,p))\right)$$
$$= p\left(\left.\frac{d}{dt}\right|_{t=0} \psi_{\exp(-t\xi)}(q)\right)$$

for all $(q, p) \in T^\vee M$.

References

[MS17] Dusa McDuff and Dietmar Salamon. *Introduction to Symplectic Topology*. 3rd. Oxford Graduate Texts in Mathematics 27. Oxford University Press, 2017.

[Sil08] Ana Cannas da Silva. *Lectures on Symplectic Geometry*. 2nd edition. Lecture Notes in Mathematics 1764. 2008.

[Wei94] Charles A. Weibel. *An introduction to homological algebra*. Vol. 38. Cambridge studies in advanced mathematics. Cambridge University Press, 1994.